

Mahmoud Abdo

Software Engineer | Full Stack (.NET & Angular)

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PROFILE

Software Engineer focused on building reliable and scalable full-stack applications. Strong foundation in Computer Science, Data Structures, Algorithms, OOP, Databases, and System Design. Experienced in developing backend services, RESTful APIs, relational databases, and modern web applications, with a focus on security, performance, maintainability, and practical engineering.

PROJECTS

E-Commerce Web API 🔄 Live API 🔗

- **Architected** a production-grade E-Commerce API using **Clean Architecture**, **DDD**, and **CQRS (MediatR)** with strict layering.
- **Optimized performance by 80%** using **Redis caching**, efficient server-side pagination and **T-SQL queries**, and **EF Core concurrency row-versioning** to ensure high scalability and safe concurrent updates.
- **Engineered** a Zero-Trust security model implementing **JWT authentication with refresh rotation**, **RBAC**, policy-based authorization, request validation, and **API rate limiting** to secure endpoints.
- **Integrated Stripe payment processing** with idempotent transaction handling and webhook validation, while orchestrating background workflows with **Hangfire** for OTP and email delivery via **MailKit**.
- **Achieved +70% code coverage** through comprehensive **xUnit/Moq** testing suites and adopted the **Result Pattern** to enforce consistent and maintainable error handling across services.
- **Implemented** scalable backend patterns including **Repository + UnitOfWork**, **FluentValidation**, **AutoMapper DTO mapping**, and caching strategies to improve maintainability and system performance.
- **Designed scalable RESTful APIs** with DTO abstraction, standardized response contracts, input validation, and consistent error handling to improve API reliability and client integration.
- **Deployed** the API to production on **MonsterASP.NET** with **HTTPS enforcement**, **Serilog structured logging**, global exception middleware, and **API versioning** ensuring backward compatibility and observability.

ExpenseTracker – Full Stack Finance Platform 🔄 Web Platform 🔗

- Built a full-stack finance platform managing income, expenses, budgets, and financial insights using **ASP.NET Core & Angular**.
- Designed secure authentication and authorization with **JWT**, password hashing, and user-specific data isolation.
- Engineered dashboard and transaction workflows with validation, business rules, pagination, and transactional consistency.
- Applied Clean Architecture with **CQRS**, **MediatR**, **Repository Pattern**, **Unit of Work**, and feature-based design.
- Developed a responsive Angular frontend with reusable components, reactive forms, financial charts, and REST API integration.
- Containerized the complete platform using **Docker Compose** with **Nginx** and **SQL Server** for consistent deployment.

DVLD Management System – Enterprise Desktop Application 🔄

- Architected a full-scale governmental Driving License & Vehicle Department system using structured **3-Layer Architecture**.
- Designed and implemented workflows including license issuance, renewal, replacement, detention, release, test scheduling.
- Built a centralized Business Layer enforcing validation rules, state transitions, and transactional consistency across all operations.
- Developed an **ADO.NET** data access layer using parameterized queries and transactions to ensure data integrity.
- Managed multi-entity lifecycle coordination (People, Drivers, Applications, Licenses, Users, Tests) with separation of concerns.
- Integrated structured logging through **Windows Event Log** for auditing, monitoring, and error traceability.
- Applied **SOLID** principles and modular design to ensure maintainability, scalability, and long-term extensibility.

TECHNICAL SKILLS

Languages: C#, C++, Python (Basics) | **Frontend:** JavaScript, TypeScript, HTML5, CSS3, Tailwind CSS, **Angular**
Backend & Frameworks: .NET, ASP.NET Core, Windows Forms, Entity Framework Core, ADO.NET, LINQ, RESTful APIs, Unit testing
Architecture & Design: Layered, Clean Architecture, CQRS (MediatR), SOLID Principles, DDD, Repository Pattern, Unit of Work
Database: SQL Server, T-SQL, Database Design, Caching (Redis)
Security: JWT Authentication, ASP.NET Core Identity, Role-Based Access Control, Abuse protection, Data Integrity
Tools: Git, GitHub, Swagger, Postman, Linux, SSMS, Visual Studio, Visual Studio Code, Docker

EDUCATION

B.Sc. in Computer Science, Mansoura University: Faculty of Computers & Information GPA: 3.2/4.0
Relevant courses: Data Structures & Algorithms, OOP, Operating Systems, Database Systems, Computer Architecture, Discrete Mathematics, Software engineering, Data communications, Web programming, Probability

LANGUAGES

Arabic (Native) | English (B2 – Upper-Intermediate) | German (A2 – Basic User)